

MD Saib Hossain

AI Engineer & Researcher

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Profile

AI Engineer & Systems Architect with expertise in Scalable Agentic AI, LLM Applications, and Multi-Agent Systems. Skilled at translating research into enterprise production, particularly developing comprehensive medical AI diagnostic tools. Proficient with engineering autonomous workflows with LangGraph, LLM APIs, and cloud deployment. Also proficient with overseeing the entire AI product lifecycle, from budgeting to workflow costs to scaling deployment into the real world.

Education

United International University (UIU) (2022 - 2026)

B.Sc. in Computer Science & Engineering

CGPA: 3.46 out of 4.00

Research & Professional Experience

1. Research Assistant | Institute of Advanced Research (IAR) (UIU) | Nov 2025 - Present

Led the development of an AI diagnostic app and clinical chatbot for rural healthcare, overseeing ethics reviews and risk mitigation. Working a deep learning pipelines for medical image analysis, optimized system latency, and authored research on AI in healthcare, Agentic AI security, and model reliability.

2. Machine Learning Engineer Intern | Fluvo Soft (Remote) | Nov 2025 - Mar 2026

Designed a synthetic face generation pipeline utilizing StyleGAN2 and Stable Diffusion XL (SDXL) to address data scarcity. Developed and deployed an Agentic RAG system to automate workflows for customer support and HR teams.

3. Volunteer Research Assistant | Ruach Tech (London, UK- Remote) Jan - Mar 2026

Conducted research on AI safety and alignment, analyzed and summarized advanced AI research papers. Supported the development of safe AI system frameworks for ongoing R&D projects.

Deployed AI Projects & Services

1. **AI Lung Cancer Screening & Clinical Chatbot:** UIU funded to deployed a web platform integrating AI-driven lung cancer screening with a specialized clinical triage chatbot for preliminary diagnostic guidance. For helping the local community and doctors. <https://lungcancerscreening.vercel.app/>

2. **Cervical Cancer Screening Prototype:** For non-govt organizations I deployed a cloud based healthcare screening tool on Render for automated cervical cancer type detection to support rapid medical triage and regional health initiatives. <https://via-prototype-gv0g.onrender.com/index.html>

3. **Educational AI Assistant:** Developed and hosted a low-latency academic chatbot for providing localized, real-time learning support specifically for Bangladeshi students. <https://phi-chat-ui.vercel.app/>

Open-Source Projects

1. MemoryOS - LLM Memory & Sandbox test.

For learning memory layer for modern LLMs and agentic systems from scratch used a local PostgreSQL database and ollama to in depth explore context storage, optimization, and retrieval. Additional, conducted targeted attack simulations within a sandbox environment to test system security and reliability. <https://github.com/Saibhossain/MemoryOS.git>

2. ScholarAgent - AI Multi-Agent Scholarship Advisor

Developed a multi-agent scholarship intelligence system using CrewAI and Ollama with (34–42s) per-agent inference latency and autonomous academic planning workflows. A CrewAI-based agentic AI pipeline for scholarship retrieval, roadmap generation, and SOP automation with real-time web search and (146s) end-to-end execution latency. <https://github.com/Saibhossain/ScholarAgent.git>

3. Self-Reflective Retrieval-Augmented Generation (Self-RAG):

Converted the latest Self-RAG research a framework where the AI adaptively retrieves external data and critiques its own responses on-demand into a production-ready prototype. This system can

maximize reliability, improve retrieval correctness, and ensure cost efficiency, designed to seamlessly scale to millions of concurrent requests. https://github.com/Saibhossain/self_rag_system.git

4. Agentic AI Technical Interviewer

Systematically make a non-linear, autonomous AI interviewing agent that dynamically parses PDF resumes to generate tailored technical assessments. Orchestrated complex, stateful conversation flows including real-time answer evaluation, clarifying hints, and adaptive follow-ups.

<https://github.com/Saibhossain/Interview-agent.git>

5. MedAssist-LLM-iPhone

Fine-tuned a lightweight *Qwen2.5-0.5B-Instruct* LLM on a medical Question-Answering dataset and optimized it for fully offline inference on an iPhone X. <https://github.com/Saibhossain/MedAssist-LLM-iPhone.git>

6. Real-Time Vision + LLM Agent “LLM with Eyes.”

Developed an autonomous vision agent using YOLOv11 and the Gemini 2.5 Flash LLM. Pipeline segments objects from a live camera feed in real-time and routes visual data to an LLM for context-aware identification. <https://github.com/Saibhossain/Eye-of-LLM.git>

Core Technical Skills

Languages: Python, C++, SQL.

ML& DL: NumPy, Pandas, Scikit-Learn, PyTorch, Vision Transformers, Deep Learning.

LLM Use: OpenAI API, Gemini API, Groq API, Ollama, AI Guardrails, Sandbox Testing, AI Governance.

Agentic AI & LLM: LangChain, LangGraph, CrewAI, RAG, Autonomous Workflows.

ML/InfOps, Eval: Langsmith, DeepEval, Docker, FastAPI, Cloudflare, Render, Supabase.

System Architecture & Technical Writing: End-to-End Pipeline Design, Product Latency Optimization, DSA, Scientific Manuscript Authorship, Peer-Review Navigation.

Project Management & Leadership: Stakeholder Collaboration, AI Product Prototyping (MVP), Cross-Functional Leadership, Technical Resource Planning.

First Author Publications

IEEE ICCIT 2025 (3 Papers):

1. Fracture Classification (ResNet, YOLO, DeepLabV3) — AUC +3.8%, mAP +4.2% [Link](#)
2. Bangla Sign Language Learning System - 98% accuracy. <https://ieeexplore.ieee.org/document/11491587>
3. Satellite-based Environmental Analysis <https://ieeexplore.ieee.org/document/11490302>

Achievements & Awards

1st Runner-up – Kaggle Competition (Deep Learning in Computational and Spatial Biology)

Best Abstract Award, same competition (Spatial Biology)

Leadership & Communication Skills

Volunteer at Alo-Bhubon Trust in healthcare & community development initiatives. (2025-Present) [Link](#)

UIU Mars Rover Team (UMRT) | Optimized autonomous navigation algorithms, contributing to real-time obstacle detection, path planning, and mission strategy for the Mars rover. (2023 - 2024) [Link](#)

General Secretary | National Children’s Task Force (NCTF) Satkhira (2014 – 2017) [Link](#)

Certifications

Data Science Fundamentals Python & SQL: <https://www.coursera.org/account/accomplishments/specialization/certificate/M0R6WZHG3OWW>

Pytorch: <https://www.coursera.org/account/accomplishments/verify/TW48IZYTPFLB>

IBM AI Engineering: <https://www.coursera.org/account/accomplishments/professional-cert/certificate/8G2L899L1IFE>

Introduction to Cybersecurity: https://www.credly.com/badges/a81366ef-a9b3-4a9f-bc1d-3b3ccdd3f3d3/public_url

Red Hat System Administration: <https://www.credly.com/badges/f76fc449-70e1-4013-81ba-975d8ae167dd>

Project Deep Agents: <https://academy.langchain.com/certificates/u0hroxniov>

AWS AI Practitioner Challenge: <https://www.udacity.com/certificate/e/f720698a-2815-11f1-83da-13cb2aea2e67>

Model Context Protocol (MCP): <https://verify.skilljar.com/c/6xy3crte3zpb>

Financial Markets: [Link](#)